

Higher Tier only questions

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		A, D and E – all three needed (1) they have equal numbers of protons and electrons (1)		2		2		
		(ii)		B and F – both needed for (1) B has a +/1/1+ charge and F has a +2/2+ – both needed for (1) award (2) for B ⁺ and F ²⁺		2		2		
	(b)			any of following for (1) - atoms having the same number of protons but different number of neutrons - same atomic number but different mass number - atoms of the same element having different number of neutrons / different mass number award (1) for comparison of ¹² C and ¹⁴ C e.g. ¹²C has 6 protons and 6 neutrons and ¹⁴C has 6 protons and 8 neutrons ¹⁴C has two more neutrons than ¹²C	2			2		
				Question 3 total	2	4	0	6	0	0